

2024 Tenth Indian Control Conference (ICC)



Proceedings

Technical Co-sponsor: IEEE Control Systems Society

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Welcome by the General Co-Chairs

We are delighted to warmly welcome all participants attending the 2024 Tenth Indian Control Conference (ICC-10) hosted at the Indian Institute of Science Education and Research Bhopal (IISER), from December 9th to 11th 2024. The inaugural ICC took place at IIT Madras in 2015, with subsequent editions being held at Mahindra Ecole Centrale-IIT Hyderabad, IIT Guwahati, IIT Kanpur, IIT Delhi, IIT Hyderabad, IIT Bombay (virtual mode), IIT Madras, and GITAM University Visakhapatnam. Being the tenth in the series, ICC-10 is a milestone for the ICC franchise.

Established in 2008, IISER Bhopal is an institution of national importance located in Bhopal. Bhopal, the City of Lakes, is a beautiful city located in the central part of Madhya Pradesh, which is called the heart of India. Bhopal serves as the capital of Madhya Pradesh and is one of the greenest cities in India.

Continuing the tradition of past years, the IEEE Control Systems Society (CSS) has graciously provided technical co-sponsorship for the ICC-10, ensuring the inclusion of the Proceedings in IEEE Xplore. We express our sincere gratitude to the CSS for their ongoing support.

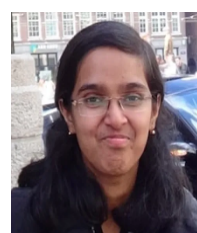
In continuation of another tradition, ICC-10 is happy to provide financial support to 69 individuals (including 65 students, 34 of whom are student co-authors of papers accepted at ICC-10) to attend ICC-10. The support covers registration, stay, and travel (partial).

Our heartfelt thanks go to the management of IISER Bhopal for graciously hosting the conference. Additionally, we extend our gratitude to our sponsors: IEEE Control System Society, TCS Research, Qualisys, Edutech, Mathworks, Delsys, Netweb Technologies, MG Automation Technologies and the State Bank of India.

We would like to acknowledge the Organizing Committee, especially the Local Organizing Committee and the dedicated student volunteers, for their invaluable contributions. Their efforts have significantly eased our responsibilities, and we take this opportunity to express our gratitude to each one of them.



M. Vidyasagar,
General Chair
Indian Institute of Technology, Hyderabad



Atreyee Kundu,
General Co-Chair
Indian Institute of Technology, Kharagpur

Report from the 2024 Tenth ICC Program Co-chairs

Welcome to the 2024 Tenth Indian Control Conference, being held from December 09-11, 2024, at the IISER Bhopal campus.

First, we would like to thank the International Program Committee for their excellent support in selecting high-quality papers. We also would like to thank the members of various committees for their contributions in organizing this event. We also thank our sponsors for their continuing support.

The conference received 161 submissions as contributed papers and 12 submissions as experimental abstracts. After a rigorous peer review process, approximately 51% of these papers got accepted for the presentation at ICC-10. The final program includes 3 plenaries, 2 industry semi-plenaries, 3 tutorial sessions, 2 industry tutorial sessions, 15 regular sessions, 1 experimental abstracts session and 4 invited talks by young and upcoming researchers. The latter is a new element we have added to ICC since the 2023 edition to help promising young researchers who are new to the Indian academia get introduced to the community at large.

The conference is for three days with the first day (9th December) reserved for tutorial sessions. The forenoon of the second day (10th December) will start with a plenary lecture, followed first by invited talks and then by 4 parallel tracks of regular sessions. To recognize experimental work and encourage more young researchers to take up experimental research, ICC-10 features a special session that enables young researchers to showcase their experimental research work. This session will be held on the forenoon of 10th December. The afternoon of the second day starts with a plenary talk followed by 2 industry semi-plenaries, and closes with 4 parallel tracks of regular sessions. The third day's (11th December) program also starts with a plenary lecture. We have 4 parallel tracks of regular sessions in the forenoon and 3 parallel tracks of regular sessions in the afternoon. Additionally, to mark the completion of 10 years of ICC, a special commemorative session has been scheduled on the afternoon of 11th December.

Over the past 10 years, the ICC has become a vibrant gathering where current and exciting research ideas are discussed. The submissions received at ICC-10 were of very high quality and represented our research community's diverse and evolving interests. These ranged from power systems, aero-mechanical systems, and robotics, to networked control and machine learning. We hope it will be a rewarding experience to attend this Tenth ICC and look forward to your continued participation in the years to come.



Ankur Kulkarni,
Program Chair
Indian Institute of Technology, Bombay



Soumya Ranjan Sahoo,
Program Co-Chair
Indian Institute of Technology, Kanpur

Plenary Sessions

Learning and Control in Countable State Spaces

Rayadurgam Srikant

University of Illinois at Urbana-Champaign

December 10, 2024, 09:30 – 10:30, VH Auditorium



Abstract: We will consider policy optimization methods in reinforcement learning where the state space is countably infinite. The motivation arises from control problems in communication networks and matching markets. Specifically, we consider the popular Natural Policy Gradient (NPG) algorithm, which has been studied in the past only under the assumptions that the cost is bounded and the state space is finite, neither of which holds for the aforementioned control problems. Assuming a Lyapunov drift condition, which is naturally satisfied in some cases and can be satisfied in other cases at a small cost in performance, we design a state-dependent step-size rule which dramatically improves the performance of NPG for our intended applications. In addition to experimentally verifying the performance improvement, we also theoretically show that the iteration complexity of NPG can be made independent of the size of the state space. The key analytical tool we use is the connection between NPG stepsizes and the solution to Poisson's equation. In particular, we provide policy-independent bounds on the solution to Poisson's equation, which are then used to guide the choice of NPG stepsizes.

Biography: Prof. Rayadurgam Srikant received his B.Tech. from the Indian Institute of Technology, Madras in 1985, his M.S. and Ph.D. from the University of Illinois in 1988 and 1991, respectively, all in Electrical Engineering. He was a Member of Technical Staff at AT&T Bell Laboratories from 1991 to 1995. He is currently with the University of Illinois at Urbana-Champaign, where he is the Fredric G. and Elizabeth H. Nearing Professor in the Department of Electrical and Computer Engineering, and a Professor in the Coordinated Science Lab. He is also the co-director of the C3.ai Digital Transformation Institute. He is the recipient of the 2015 INFOCOM Achievement Award and the 2019 IEEE Koji Kobayashi Computers and Communications Award, and has received several Best Paper awards including the 2015 INFOCOM Best Paper Award and the 2017 Applied Probability Society Best Publication Award. He was the Editor-in-Chief of the IEEE/ACM Transactions on Networking from 2013 to 2017. His research interests include communication networks, machine learning, and applied probability. He is a Fellow of the IEEE.

Linear Quantum Control Engineering

Ian Petersen

Australian National University

December 10, 2024, 14:15 – 15:15, VH Auditorium



Abstract: This lecture will present a collection of results in the area of linear quantum control engineering. It will discuss models for quantum systems using the Heisenberg pictures of quantum mechanics to describe continuous linear quantum systems. It will discuss closed loop approaches to quantum control involving coherent quantum feedback control in which the controller is also a quantum system. It will discuss quantum H_∞ control, quantum LQG control, coherent quantum observers and coherent quantum state estimation. The lecture will also cover the quantum Kalman decomposition as well as recent results on quantum risk sensitive control and networks of quantum linear systems as quantum memories. Applications in the areas of quantum optics and quantum electromechanical systems will also be presented.

Biography: Prof. Ian R. Petersen was born in Victoria, Australia. He received a Ph.D in Electrical Engineering in 1984 from the University of Rochester. From 1983 to 1985 he was a Postdoctoral Fellow at the Australian National University. From 2017 he has been a Professor at the Australian National University in the School of Engineering. He was the Interim Director of the School of Engineering at the Australian National University from 2018-2019. From 1985 until 2016 he was with UNSW Canberra where he was a Scientia Professor and an Australian Research Council Laureate Fellow in the School of Engineering and Information Technology. He has previously been ARC Executive Director for Mathematics Information and Communications, Acting Deputy Vice-Chancellor Research for UNSW and an Australian Federation Fellow. He has served as an Associate Editor for the IEEE Transactions on Automatic Control, Systems and Control Letters, Automatica, IEEE Transactions on Control Systems Technology and SIAM Journal on Control and Optimization. He also served as an Editor for Automatica. He is a fellow of IFAC, the Asian Control Association, the IEEE and the Australian Academy of Science. His main research interests are in robust control theory, quantum control theory and stochastic control theory.

Machine Learning Applications in Biology

Sanghamitra Bandyopadhyay

Indian Statistical Institute, Kolkata

December 11, 2024, 09:00 – 10:00, VH Auditorium



Abstract: Machine Learning (ML) algorithms are finding extensive applications in all domains of scientific research, in particular in diverse areas of biology for making novel discoveries and for gaining deeper insights into various processes of life. In this talk we will first present brief overviews of artificial intelligence, machine learning and molecular biology. In particular, a quick introduction to the central dogma of molecular biology will be provided, which is fundamental to the understanding of a large class of machine learning applications in biology. Various kinds of molecular biology data sets will be mentioned. We will then discuss some case studies, namely, molecular target prediction, a graph theoretic method for biomarker identification and

graph-based approaches in drug interaction studies. The talk will conclude with a mention of some issues and challenges in this area.

Biography: Prof. Sanghamitra Bandyopadhyay did her B Tech, M Tech and Ph. D. in Computer Science from Calcutta University, IIT Kharagpur and Indian Statistical Institute, respectively. She then joined the Indian Statistical Institute as a faculty member, and became the Director in 2015. Since 2020, she is continuing her second tenure as the Director of the Institute. Her research interests include computational biology, soft and evolutionary computation, artificial intelligence and machine learning. She has authored/co-authored several books and numerous articles in journals, book chapters, and conference proceedings and has a citation h-index of 62. Prof. Bandyopadhyay has worked in many Institutes and Universities worldwide. She is the recipient of several awards including the Shanti Swarup Bhatnagar Prize in Engineering Science, TWAS Prize, Infosys Prize, JC Bose Fellowship, Swarnajayanti fellowship, INAE Silver Jubilee award, INAE Woman Engineer of the Year award (academia), IIT Kharagpur Distinguished Alumni Award, Humboldt Fellowship from Germany, Senior Associateship of ICTP, Italy, young engineer/scientist awards from INSA, INAE and ISCA, and Dr. Shanker Dayal Sharma Gold Medal and Institute Silver from IIT, Kharagpur, India. She is a Fellow of all the Science Academies and the Engineering Academy of India, IEEE, The World Academy of Sciences (TWAS), and International Association for Pattern Recognition (IAPR). She serves as a member of the Science, Technology and Innovation Advisory Council of the Prime Minister of India (PM-STIAC). In 2022, she was conferred the Padma Shri award of the Government of India.

Industry Semi-Plenary

Stability-Theory Perspective into Optimization: Design and Analysis of Algorithms

Mayank Baranwal

TCS Research

December 10, 2024, 15:30-16:30, TuET1.1, VH Auditorium

Abstract: Optimization algorithms are the driving force behind many machine learning formulations, and recent progress in this field has been greatly aided by a better understanding of these algorithms and their implementation for specific machine-learning problems. However, the field lacks a generalized tool to design and analyze optimization algorithms, making it non-intuitive for practitioners to develop their accelerated variants. The study of gradient flow and its connection to dynamical systems has a long history in mathematics. This talk aims to further explore this relationship from a continuous-time perspective. In particular, we introduce a generalized framework for designing accelerated optimization algorithms based on recent advances in the fixed-time stability theory of continuous-time dynamical systems. This framework allows for the strongest possible convergence guarantees and easily extends to a subclass of non-convex functions. This talk is part of a larger effort to connect the fields of dynamical systems and optimization and to advance our understanding of the underlying mathematical principles and their applications.

Biography: Mayank Baranwal is a Senior Scientist with the research division of Tata Consultancy Services (TCS) in Mumbai. He also holds an Adjunct appointment with the Systems and Control group at the Indian Institute of Technology, Bombay. Prior to joining TCS, he was a postdoctoral scholar in the Department of Electrical and Computer Engineering at the University of Michigan, Ann Arbor. He received his Bachelors in Mechanical Engineering in 2011 from Indian Institute of Technology, Kanpur, and MS in Mechanical Science and Engineering in 2014, MS in Mathematics in 2015 and PhD in Mechanical Science and Engineering in 2018, all from the University of Illinois at Urbana-Champaign. His research interests are in modeling, optimization, control and inference in network systems with applications to distributed optimization, supply-chain network, power networks, control of microgrids, bioinformatics and computational biology, and deep learning theory. Mayank is a recipient of the Institute Silver Medal in 2011 (from IIT Kanpur), the ME Outstanding Publication Award in 2017 (from the University of Illinois), and the Young Scientist Award in 2022 (Tata Consultancy Services). He is also a young associate of the INAE.

Control Challenges in SDVs: Zonal Architecture, SOFAEE, and AI/ML

Integration

Muralikrishnan Elumalai

Edutech

December 10, 2024, 15:30 – 16:30, TuET2.1, Seminar Hall 1

Abstract: Software-defined vehicles (SDVs) are redefining automotive technology. They integrate modular architectures like SOFAEE and Autoware with emerging zonal designs to enhance scalability and control. These architectures optimize resource utilization and decentralize subsystems, addressing fault tolerance and real-time processing challenges. The integration of AI/ML introduces new dimensions for perception, decision-making, and control, but also demands robust strategies for real-time optimization and cybersecurity. With SDVs increasingly connected, safeguarding against cyber threats while ensuring functional safety has become critical. This presentation explores control strategies, system-level challenges, and collaborative solutions necessary for advancing SDV deployment. It highlights the importance of adaptive control frameworks, open architectures, and standardized protocols to address these complexities. By bridging control theory with emerging technologies, this discussion provides actionable insights for researchers and practitioners to accelerate innovation in autonomous mobility.

Biography: Muralikrishnan serves as the Head of Autonomous and e-Mobility at Edutech, a pioneering organization with over 30 years of legacy in transforming education and technology. With a focus on cutting-edge fields like Autonomous Systems, Unmanned Ground Vehicles (UGVs), Electric Vehicle (EV) research, and Software-Defined Vehicle (SDV) architectures, Muralikrishnan has successfully implemented end-to-end autonomous solutions for both industries and academic institutions. His work extends to the forefront of motion capture systems and 5G research and beyond, where he combines his deep technical expertise with strategic insights to address complex challenges. Muralikrishnan is a passionate advocate for bridging the gap between academia and real-world applications, enabling innovative solutions that drive progress in autonomous and mobility systems. Academically, Muralikrishnan holds a Master of Science in Telecommunications Engineering from Victoria University and a Bachelor of Engineering in Electronics and Communication Engineering from the University of Madras. His career reflects a seamless integration of technical innovation, academic excellence, and leadership, underscoring his role as an expert in the field of autonomous technologies and future mobility.

Invited Talks

Efficient Conflict-Free Path Planning for Multi-Agent Systems: Methods and Applications

Sharad Kumar Singh, IIT Indore

December 10, 2024, 10:50- 11:20, TuBT1, VH Auditorium

Abstract: In a network of multiple agents, the behaviours of friendship and hostility arise naturally. This leads to signed networks in which the edge weights can take even negative values. Structural balance is an important property associated with such networks which stems from a certain bipartiteness of the networks. Under the Laplacian flow, signed networks exhibit richer dynamics as opposed to networks with strictly positive edge weights. The focus of the talk will be on the role of structural balance in achieving these behaviours of consensus or bipartite consensus in signed networks.

Biography: Dr. Sharad Kumar Singh is an Assistant Professor in the Department of Electrical Engineering at IIT Indore. His research interests include game theory, control and optimization, robotics, and multi-agent systems. Dr. Singh earned both his M.S. (by research) and Ph.D. in Electrical Engineering from IIT Madras. Following his Ph.D., he worked as a Researcher in Mobile Robotics at Adverb Technologies in Noida. Later, he served as an Operations Research Analyst at Solver Minds Solutions and Technologies in Chennai. Dr. Singh's diverse professional experience and strong academic background uniquely position him to contribute to advancements in multi-agent systems, robotics, and game theory.

Multi-agent Systems – Cohesive Control, its Applications, and Effects of Delay

Anuj Tiwari, IIT Madras

December 10, 2024, 10:50- 11:20, TuBT2, Seminar Hall 1

Abstract: Cohesion in networks during transitions from one consensus value to another, i.e., the ability of agents to respond in a similar manner during the transition, can be as important as achieving the new consensus value. Existing decentralized network control strategies are mainly concerned with the speed of convergence to the final consensus value. However, even with increased convergence speed, the level of cohesion loss during transitions can be large. This loss of cohesion during transition (and tracking of varying consensus values) can be alleviated using a recently developed delayed self-reinforcement (DSR) approach. However, the current DSR-based approach assumes ideal conditions with agents having instant access to neighbor information – without network delays arising during sensing or communication between neighbors, as well as computation of control actions of each agent, which can cause instability. In this talk, I will present my work on addressing the issue of cohesive transitions in networks with delays. Additionally, I will discuss the results of cohesion control in formation control of multi-agent systems, multi-robot flexible object transport and inter-vehicle spacing control in connected vehicles.

Biography: Anuj Tiwari completed his B.Tech. degree from IIT Guwahati, India, in June 2017 and the Ph.D. degree at University of Washington, Seattle WA, USA, in December 2022. After a short stint as a Postdoctoral Scholar in Boeing Advanced Research Center till June 2023, he has joined IIT Madras as an Assistant Professor in the Mechanical Engineering Department. His research focus is on distributed control of networked multi-agent systems, with applications in robotics for advanced manufacturing, intelligent transportation systems, and cyber physical systems.

Data-based Stability Analysis of Strongly Autonomous nD Systems

Mousumi Mukherjee

IEST Shibpur

December 10, 2024, 10:50- 11:20, TuBT3, Seminar Hall 2

Abstract: With the advances in sensing technology, computing capabilities, and storage infrastructure there has been a surge in using data for an increasing number of applications. However, the vast literature on data-based analysis and control is mostly focused on dynamical systems that evolve over one independent variable, namely, time. In this talk, we shall discuss about data-based conditions for checking the stability of a special class of systems having more than one independent variable. It is known that because of the lack of natural ordering of the domain, the problem of defining and studying the stability of systems having n independent variables (nD systems) is not straightforward. We focus on the following two notions of stability: (i) stability with respect to a given direction, and more generally, (ii) conic stability, that is, stability with respect to subsets of the domain having the structure of a cone, and provide data-based tests for checking them.

Biography: Mousumi Mukherjee is currently an Assistant Professor in the Department of Electrical Engineering at the Indian Institute of Engineering Science and Technology (IEST), Shibpur. Prior to joining IEST, Shibpur, she served as a DST-INSPIRE Faculty at IIT Madras. She obtained her PhD from the Control and Computing group of the Electrical Engineering Department at IIT Bombay, where her research earned the “Excellence in PhD” award. Following her doctoral studies, she was a postdoctoral researcher in the Chair for Mechatronics at the Technical University of Kaiserslautern, Germany. Her research interests encompass systems and control theory, with a particular focus on algebraic analysis of dynamical systems, multidimensional (nD) systems, computational commutative algebra, and data-driven control.

Constraint Motion Planning of Non-Holonomic Vehicles

Bhargav Jha

IIT Kharagpur

December 10, 2024, 10:50- 11:20, TuBT4, Board Room 1

Abstract: Moving platforms such as ships with firing turrets, mobile air-defense systems, surveillance aircraft, and military vehicles with directed sensors are all examples of moving pursuers equipped with rotating platforms. The platforms can be a turret, a directed energy weapon, or a gimbaled camera and sensor. This talk will explore time-optimal and energy-optimal motion planning strategies for such vehicles. We will begin by introducing a novel vehicle model, called the Dubins-Laser system. Following this, we will discuss how first-order necessary optimality conditions can be leveraged to reduce the trajectory search space to a finite set of candidate solutions. Finally, numerical simulations will be presented to demonstrate the practical applications of the proposed analytical results.

Biography: Bhargav Jha is currently an Assistant Professor at the Department of Electrical Engineering, IIT Kharagpur. Prior to this he was a post-doctoral fellow with the Electrical and Computer Engineering Department at the Michigan State University, USA. He received his B.E. (Hons) degree in Electronics and Instrumentation from Birla Institute of Technology & Science Pilani, India in 2016 and the Masters as well as Ph.D. degree in Aerospace Engineering from Technion-Israel Institute of Technology in 2019 and 2022, respectively. His current research interests lie at the intersection of motion planning, optimal control, and differential game theory.

Tutorials

Risk-Sensitive Reinforcement Learning Via Policy Gradient Search

Prashanth L. A., IIT Madras

December 9, 2024, 09:00-11:45, MoAT1.1, Seminar Hall 2

Abstract: The objective in a traditional reinforcement learning (RL) problem is to find a policy that optimizes the expected value of a performance metric such as the infinite-horizon cumulative discounted or long-run average cost/reward. In practice, optimizing the expected value alone may not be satisfactory, in that it may be desirable to incorporate the notion of risk into the optimization problem formulation, either in the objective or as a constraint. In this tutorial, we consider risk-sensitive RL in two settings: one where the goal is to find a policy that optimizes the usual expected value objective while ensuring that a risk constraint is satisfied, and the other where the risk measure is the objective. Various risk measures to be considered include exponential utility, variance, percentile performance, chance constraints, value at risk (quantiles), conditional value-at-risk, prospect theory, and general coherent risk measures. We survey some of the recent work in this area specifically where policy gradient search is the solution approach. In the first risk-sensitive RL setting, we present a template for policy gradient-based risk-sensitive RL algorithms using a Lagrangian formulation. For the setting where risk is incorporated directly into the objective function, we present policy gradient algorithms for an exponential utility formulation, cumulative prospect theory, and coherent risk measures.

Introduction to Data-Driven Control

Vaibhav Katewa, IISc Bangalore

December 9, 2024, 09:00-11:45, MoAT3.1, Seminar Hall 1

Abstract: Recently, there has been a renewed interest in addressing traditional control problems without the knowledge of the system model. In such cases, the data collected from the system is used to develop the controllers. Although related to the traditional system identification problem, this setting is distinct in the sense that direct methods are used that provide the controller without identifying the system model explicitly. In this tutorial, we will cover the basics of data-driven control design for canonical deterministic control problems like feedback control, stabilization, LQR, LQG etc. Further, we will also look at the stochastic setting with finite sample guarantees for system identification and control design in the presence of random noises in the data.

Biomolecular Systems: Decision-Making and Memory

Vinod Kumar, Rudra Prakash, and Shaunak Sen, IIT Delhi

December 9, 2024, 14:45 – 17:30, MoCT1.1, VH Auditorium

Abstract:

Decision-making and memory are widespread behaviours in biomolecular systems. Because the underlying principles are nonlinearity and positive feedback, these topics are directly linked to control engineering. This tutorial elaborates on these topics. First, benchmark models, mathematical and experimental, would be covered, to convey examples of these behaviours. Second, basic methods of Interval Analysis would be covered to provide a natural and rigorous tool to solve associated mathematical equations. Third, advanced topics arising from the study of these behaviours, such as in Algebraic Geometry, Monotone Dynamical Systems, and Chemical Reaction Network Theory would be discussed, to give a sense of open problems. This tutorial would add to the control engineering perspective on decisions and memory.

Industry Tutorials

Introduction to Model Predictive Control (MPC) with MATLAB and Simulink

Pranav Lad and Dhruv Chandel

Mathworks

December 9, 2024, 11:45-13:15, MoBT1.1, Seminar Hall 1

Abstract: Model Predictive Control (MPC) is an advanced control technique that has been used for process control since the 1980s. With the increasing computing power of microprocessors as well as high-speed optimization algorithms, the use of MPC has spread to many real-time embedded applications, often used in automotive, aerospace, industrial automation, and other industries. MPC can handle multi-input multi-output (MIMO) systems with coupled input-output channels as well as constraints on inputs, outputs, and states, which are challenging to handle with classical control methods. This workshop delves into the principles and applications of MPC, providing participants with a comprehensive understanding of the concepts and practical implementations in the simulation environment. During the workshop, we will discuss what is MPC, how it works, various categories of MPC, and their implementation. Leveraging the powerful computational tools of MATLAB and Simulink, participants will gain hands-on experience in designing and simulating MPC controllers for linear and nonlinear applications. Extensive libraries of MATLAB and Simulink's dynamic simulation environment will be utilized to demonstrate real-world applications and facilitate an intuitive understanding of complex control scenarios. By the end of this workshop, participants will be equipped with the skills to implement MPC in various engineering contexts, enhancing their ability to optimize performance and efficiency in control systems.

Leveraging Neural Interfaces to Enhance Control of Advanced Machines Using Delsys Technology

Nicholas Ducey and Bhawna Shiwani

Delsys Inc.

December 9, 2024, 11:45-13:15, MoBT2.1, Seminar Hall 2

Abstract: As advanced machines become more commonplace in our daily lives, it is becoming increasingly important for these machines to interact seamlessly with the humans that leverage their abilities. Delsys provides cutting-edge EMG solutions that enable engineers to design myoelectrically-controlled Human-Machine Interfaces, simplifying the complexity of human movement into intuitive control commands. This session will highlight the latest advancements offered by Delsys, including API integration and dynamic multi-channel EMG detection, which researchers can leverage to enhance their Human-Machine Interfaces. Mr. Nicholas Ducey will introduce Delsys, provide an overview of Delsys' history in HMI, review the latest advanced Delsys tools, and discuss the future of EMG in HMI. Ms. Bhawna Shiwani will present on leveraging real-time neural interfaces and dynamic motor unit identification for enhanced control of upper extremity prosthetics.

Conference Information

Awards Ceremony, Conference Dinner, and Reception

Opening Ceremony	09:00-09:20, Tuesday, December 10	VH Auditorium, IISER Bhopal
Cultural Program	20:00-21:00, Tuesday, December 10	Noor-us-Sabah Palace
Conference Dinner	21:00, Tuesday, December 10	Noor-us-Sabah Palace

The opening ceremony of ICC-10 will be at the VH auditorium of IISER Bhopal. On the same day, we will have cultural program by Madhya Laya followed by the conference dinner at Noor-us-Sabah Palace.

Madhya Laya will present Indian classical fusion music by a group of young percussionists creating a magic of percussions on various instruments. The performance will be a dialogue between these performers innovating a new dimension in the field of Tala Vadya. The group uses both folk and classical instrumental. Madhya Laya creates new aspects of Tala Vadhya Kachahri where instruments craft innovative rhythms and sounds by flouting with each other.

Noor-us-Sabah Palace, located in Bhopal, Madhya Pradesh, is a stunning example of royal heritage and Indo-Islamic architecture. Built-in the 1920s by Jahan Begum Abida Sultan, the eldest daughter of H.H. Hamidullah Khan, it was originally designed as a royal residence. The name "Noor-us-Sabah," meaning "Light of Dawn," reflects the palace's luminous charm and elegance. Perched on a hilltop, it offers a spectacular view of the serene Upper Lake, or Bhojtal, one of India's largest artificial lakes. In 1998, it was transformed into a heritage hotel while ensuring that the palace remains a living monument, accessible to visitors while retaining its original charm and architectural integrity. Its location, overlooking the Upper Lake, emphasizes its connection to the city's natural beauty and ecological balance.

Conference Registration

Registration is mandatory for all conference participants. Registration kits for all advance registrations may be picked up at the 2024 Tenth ICC Registration Desk. The Registration Desk will be open from 08:00 to 17:00 on December 9 and 10, and from 08:00 to 11:00 on December 11, except during the plenary sessions.

All registration categories include the conference proceedings (a link will be provided on the conference website from where registered participants will be able to download the proceedings), admission to the tutorial sessions and technical sessions, lunch on all three days, and one ticket for the conference dinner.

Registration Category	Conference Registration Fee (including 18% GST)*	
	Within India (INR)	Outside India (USD)
Students	6000	80
Others	15000	200

*Transaction fees may apply

Best Student Paper Award Finalists

Seventeen nominations were received for the 2024 Tenth ICC Best Student Paper Award. A panel of 4 reviewers screened the nominated papers based on originality, clarity, technical correctness, potential impact, and strength of the nomination letter. The finalists for the award are shown below in chronological order of their presentation times.

Paper Title	Student Finalist and co-author(s)	Presentation Slot
Feedback-Based Quantum Control of Nuclei Spin in Ensembles	Vishesh Kaushik and Naveen Khaneja	11:30-11:50, Control Applications I
Time Delayed Control for Manoeuvring of Magnetic Micro/Nanorobots in Drug Delivering Applications	Neeta Yadav , Abhilash Patel and Bidyadhar Subudhi	12:10-12:30, Control Applications I
Funnel-Based Control for Reach-Avoid-Stay Specifications	Ratnangshu Das and Pushpak Jagtap	12:50-13:10, Nonlinear Systems I

A second panel of four reviewers assessed the three finalists on presentation quality, delivery, significance and novelty. The winner of the 2024 Tenth ICC ICC Best Student Paper Award was **Ratnangshu Das** for his paper titled “Funnel-Based Control for Reach-Avoid-Stay Specifications” co-authored with Pushpak Jagtap.

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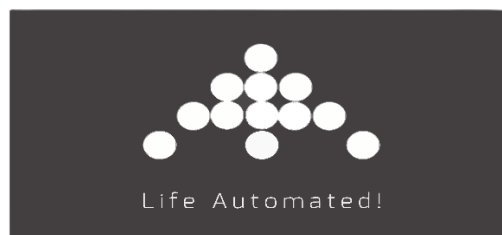
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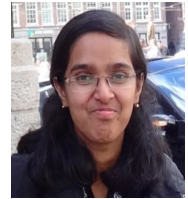
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